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A Customized Business Intelligence Dashboard Utilizing Building Information Modeling for Better Control and Management of Construction Projects

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Abstract

The construction sector is one of the primary areas that underpin a country's economic development. However, this sector is characterized by various types of obstacles, including the participation of numerous stakeholders, strict schedules, limited resources, and the management of vast amounts of data throughout the project lifecycle. Building Information Modeling (BIM) has emerged as a promising technology for centralizing and managing construction data throughout the project lifecycle. However, having the ability to extract real-time, decision-oriented insights from BIM models remains a challenge for project stakeholders. To address this limitation, this research paper explores the integration of BIM with Business Intelligence (BI) to enhance control and management of construction projects throughout the development of a customized Power BI dashboard. The proposed framework of the paper utilizes BIM's data-rich environment and Power BI's advanced analytical and visualization capabilities to deliver real-time and interactive insights about project performance and progress. The customized dashboard enables stakeholders, especially project managers, to monitor key performance indicators of the project that are related to cost and schedule. It also supports progress tracking, early identification of inefficiencies, and data-driven decision-making. To demonstrate the practical application of the proposed framework, a case study was conducted. The results indicate that integrating BIM with BI helps in enhancing project control, improving transparency, and facilitating collaboration between stakeholders through a centralized cloud platform that can be easily accessed through desktop and mobile devices.

Keywords: building information modeling; construction management; dashboard; business intelligence; digital construction; project control

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1. Introduction

1.1. Background

The construction industry plays a significant role in the economic development of countries worldwide. A recent study indicates that it represents approximately 9% of the global gross domestic product [1]. The study also shows that the global spending in the construction industry was estimated at USD 11 trillion in 2017 and is projected to reach USD 14 trillion by 2025 [1]. Despite its economic importance, this industry continues to suffer from low productivity due to the inherent complexity of construction projects. Construction projects are characterized by the involvement of multiple phases ranging

from design and planning to execution, monitoring, and control. These phases require close collaboration and coordination among diverse stakeholders including owners, engineers, architects, and contractors. Ineffective management of these entities often leads to poor project performance. Furthermore, limited transparency across project phases and difficulties in monitoring real-time progress contribute to hindering informed decision-making, increasing the probability of uncertainty, and elevating project risks. A previous study indicates that nearly 60% of construction project complexity is attributed to poor management practices, while 16% results from inadequate communication between various stakeholders [2]. These challenges have motivated firms engaged in the construction industry to seek innovative, technology-driven solutions.

The construction industry is continuously changing with technology advancing towards data-driven methods. Traditional methods are being replaced by new ones due to the improved utilization of data analytics, machine learning and other advanced technologies. A key part of this transformation has been the use of dashboards and intelligent analytical tools for effective data management. One prominent example is Microsoft Power BI. It is defined as "a business analytics platform developed by Microsoft that specializes in data connectivity, visualization, and analysis" [3]. Power BI enables organizations to gain valuable insights from massive datasets through its powerful data modeling, reporting, and interactive dashboard creation capacities. This, in turn, ensures that stakeholders make well-informed decisions, supports the development of better project management strategies, and contributes to enhanced productivity of the construction project.

1.2. Research Objectives

The main objectives of this research are as follows:

(1) Study the status of Business Intelligence in the construction sector.
(2) Design and develop a customized Power BI dashboard that utilizes Building Information Modeling for better monitoring and control of construction projects.
(3) Leveraging Power Bl capabilities for analysis and visualization of data related to construction projects.
(4) Provide a practical framework for industry adoption.

1.3. Research Roadmap

The steps involved in accomplishing the study's goals are shown below:

- A comprehensive review of the importance of the construction industry, complexity of construction projects, status of the construction sector in Saudi Arabia, BIM in construction management, Power BI dashboards, and recent studies related to BI in the construction sector.
- Set a suitable solution.
- Create straightforward guidelines for implementing the appropriate solution.

1.4. Paper Organization

The remainder of this paper is structured as follows. Section 2 provides a concise overview of the background topics including the importance and challenges of the construction industry, along with the associated literature pertinent to this research. Section 3 introduces the proposed framework to build and customize the Power BI dashboard for better control and management of construction projects. Section 4 explains the results and discusses the capabilities of the customized Power BI dashboard. Section 5 summarizes the research conclusions and provides recommendations for future work.

2. Literature Review

2.1. Importance of Construction Industry

The construction industry represents one of the most critical sectors that provide substantial contributions to national economic growth and development. It is responsible for delivering residential and non-residential infrastructure that forms the foundation of modern societies, including housing, commercial buildings, transportation networks, and public facilities. According to financial reports, construction industry contributes approximately 13% to the global gross domestic product [4]. The reports also show that global spending in the construction sector was calculated to be $11.5 trillion in 2018, and it is estimated to increase to $19.2 trillion in 2035 [4]. This massive growth in the construction sector will generate substantial employment opportunities and establish the foundation for other industries such as manufacturing, logistics, and services [5]. That means new job opportunities will be provided for a range of laborers, ranging from unskilled laborers to highly specialized professions. This employment is critical for economic stability, especially in developing countries. In addition to its economic contribution, the construction industry plays a central role in promoting sustainable development. The adoption of sustainable construction practices and innovative technological approaches has been shown to reduce operational costs and enhance energy efficiency [6]. Furthermore, the construction sector significantly contributes to improving quality of life, where it has been found that the construction of schools, hospitals, roads, and other vital infrastructure directly impacts the health, education, and well-being of the population. For instance, a study was conducted in Malawi in 2024. This study adopted the approach of studying the impact of using sustainable construction projects in the construction sector. The findings of this study show that there were significant improvements in overall quality of life and economic growth [7]. Similarly, another study was carried out in Kuwait in 2025. The study's objective was to investigate the effects of using BIM technologies on construction projects. The study's findings showed a notable increase in environmental sustainability and overall efficiency practices [8]. These findings collectively reinforce the strategic importance of the construction sector in supporting economic, social, and environmental development.

2.2. Complexity of Construction Projects

The construction sector is widely recognized for its complexity, which requires close collaboration between different stakeholders, coordination of numerous tasks, and the management of multiple technical and organizational challenges. Such complexity has been raised from the requirement of quality control, meeting tight deadlines, working within budgeted costs, and the need to comply with regulatory and safety standards. Due to its nature, construction organizations continuously seek innovative approaches to improve project management practices. In 2018, a study was conducted to investigate the various factors contributing to construction management complexity. The study's findings show that organizational factors account for approximately 70% of overall project complexity [9]. Similarly, another research paper published on 23 May 2022 reported that nearly 60% of complexity in construction projects stems from poor management, while 16% stems from poor communication [2]. Recent papers have further indicated that one of the most critical barriers to effective construction management is the slow adoption and integration of emerging digital technologies [10]. To address these challenges, continued research is essential to examine new and evolving technologies, underscore their potential value, and establish viable frameworks for their integration into the construction industry.

2.3. The Construction Sector in Saudi Arabia

The construction industry in Saudi Arabia has experienced substantial growth in recent years, driven by the implementation of large-scale infrastructure and mega projects. This expansion aligns with the national development strategy outlined in Saudi Vision 2030, which aims to diversify the economy and reduce reliance on oil revenues. The construction sector plays a central role in supporting infrastructure development, accelerating urbanization, and expanding the tourism industry. It has been reported that the population in Saudi Arabia is rising at a rate of 2.5% yearly, with only 24% of the local population owning their primary residence [11]. The same study estimated that by 2020, the Kingdom would require the construction of approximately 2.32 million new housing units to meet demand. This will open great opportunities for the construction sector, improving the overall economy and contributing to the Vision 2030 project. The construction sector in Saudi Arabia accounts for 43% of the total national investment, with residential buildings representing nearly 90% of the total projects [12]. The growth of the construction sector has created countless employment possibilities for Saudi citizens, resulting in lowering unemployment rates, improving workforce skills, and paving the way for a promising future. The spending in the construction sector in the Kingdom of Saudi Arabia in 2023 was estimated to be $65.58 billion, with an expected increment rate of 2.75%, to reach $75.12 billion in 2028 [13]. The study also states that the construction of buildings increased by 3.2% in 2022. This growth contributes to supporting the expansion of the residential sector and achieves the goals set in the vision to increase the ownership percentage to 70% by 2030.

Regardless of the substantial advancements achieved in the construction sector, the Kingdom of Saudi Arabia's construction industry continues to struggle with multiple challenges that impede its growth, operation, and efficiency. Recent research has identified project delays and cost overruns as among the most critical issues confronting the sector [14], with key contributing factors including delays in the delivery of essential materials, subcontractor failure to meet completion schedules, and the impact of unstudied inflation. Additionally, inefficient project management practices have also been identified as a major obstacle to successful project execution [15], attributed to poor project planning, shortage of skilled project managers, and insufficient coordination and collaboration among various stakeholders. Finally, the hesitation to adopt new technologies and methodologies, such as lean construction and Building Information Modeling, and reliance on traditional ones, have been identified as some of the factors that impede the efficiency of the construction sector in the Kingdom [16]. As the construction industry continues to evolve, embracing innovative technologies will be the key to staying competitive in the market.

To address these challenges, numerous research studies have been conducted and proposed recommendations tailored to the Saudi construction context [15,17,18]. The adoption of BIM has been widely recommended to improve coordination, reduce design errors, and enhance collaboration among project stakeholders [17]. In certain instances, BIM implementation has reduced claims by 90% [17]. Furthermore, utilizing Industry 4.0 technologies such as the usage of smart devices, Internet of Things, and artificial intelligence has shown promising solutions for construction challenges [18]. Studies have also demonstrated that utilizing project management software like Microsoft Project and Primavera has proven to improve project management practices [15].

2.4. Building Information Modeling in Construction Management

The construction industry continues to face persistent challenges that hinder project success such as project delays, cost overruns, and inefficient communication between stakeholders. To address these challenges, BIM has emerged as a transformative digital

technology that integrates design, scheduling, cost estimation, and operations within a shared virtual environment [19]. BIM is defined as a digital methodology that represents the building process and supports the seamless exchange and interoperability of information among project stakeholders [20]. This technology facilitates collaboration and coordination among architects, engineers, and contractors by offering a unified model that allows all permitted stakeholders to access and update.

BIM has demonstrated significant potential in improving collaboration through real-time data sharing and advanced project visualization, thereby enhancing coordination and overall project efficiency [21]. It also supports decision-making across all project phases, from conceptual design to facility management. Moreover, it provides multi-dimensional modeling (3D–7D) capabilities, allowing it to link geometry, time, cost, sustainability, and facility management data together to facilitate better planning and control of construction projects [22]. Furthermore, BIM has also been recognized as a key driver for improving the productivity of construction projects. The BIM process typically encompasses several key stages across the project lifecycle. These stages include project programming, conceptual and detailed design, performance analysis, documentation, fabrication, construction, site logistics, operation and maintenance, and finally renovation or demolition [23]. Throughout these stages, BIM supports improved design coordination, conflict detection, prefabrication accuracy, and lifecycle data management.

BIM in construction management is generally operationalized through cloud-based platforms such as Autodesk BIM 360. This platform plays a central role in document management, model coordination, issue tracking, and collaboration among various stakeholders. However, its functionality primarily focuses on information management and project coordination rather than advanced data analytics, discovering insights, and supporting well-informed decisions in real time. To overcome these limitations, BIM needs to be integrated with data-driven technologies, such as Business Intelligence tools. One of the leading tools in this field is Power BI. It is a Business Intelligence platform designed for advanced data analytics, visualization, and generation of interactive reports. It facilitates the creation of dynamic dashboards that assists project managers in tracking key performance indicators (KPIs) of the project, monitoring progress, and predicting potential issues before escalating into conflicts. The integration of Power BI with BIM platforms represents a shift from model-centric coordination toward data-driven construction management.

2.5. Power BI Dashboard

Business Intelligence is a process driven by technology that analyses structured and unstructured data to provide actionable insights that help executives make better informed decisions [24]. BI tools can track project performance, financial metrics, and operational efficiency, thereby enabling project managers to identify issues, and optimize resources in real time. Given the fast-paced and data-intensive nature of modern construction projects, timely and accurate reporting has become essential for effective project management. Among contemporary BI platforms, Microsoft Power BI has emerged as a widely adopted solution for data visualization and analytics. Power BI can be defined as a business analytical platform developed by Microsoft that provides business analytical capabilities, interactive reports and visualizations from data sources [25]. It allows users to integrate data from different sources to share insights and build interactive dashboards tailored to specific project requirements. There are many benefits for using Power BI such as extensive data connectivity. Power BI provides the capability to integrate data from a wide variety of sources, including spreadsheets, databases, cloud services, and application programming interfaces (APIs) [26]. Additionally, the platform offers customizable dashboards and cloud-based sharing features that support collaborative decision-making

among project stakeholders. The cloud platform has role-based access which controls the level of accessibility of each stakeholder based on their responsibilities.

2.6. Key Studies of Business Intelligence in Construction Sector

Recent studies have explored the application of Business Intelligence and data visualization techniques within the construction sector. Rodrigues et al. (2022) [27] investigated the relation between construction projects and data visualization, emphasizing that construction projects inherently generate vast amounts of data throughout their lifecycle. Providing stakeholders such as owners, supervisors, managers, and technicians with immediate access to these data is essential for conducting timely analyses and supporting effective decision-making. However, since construction projects consist of large amounts of data, diverse tasks and lacking the necessary tools to visualize them often, this makes it hard for construction managers to make a reliable decision. To address this issue, Rodrigues et al. proposed a methodology combining data modeling techniques with visualization tools. Their approach involved extracting data from models to Excel sheets then importing these spreadsheets into Power BI for analysis. The findings indicated that this methodology provides a collaborative environment for data analysis, allowing authorized stakeholders to access, share, and update information in real time. Moreover, the study reported improvements in decision reliability, schedule adherence, and cost control, while also contributing to sustainability objectives through more efficient resource utilization [27]. Despite these contributions, the study presented notable limitations. Model data could not be directly integrated into the Power BI platform and required intermediate export to spreadsheet format, which may introduce data inconsistency and reduce automation. Furthermore, the methodology did not support the integration or visualization of 3D BIM models within the Power BI platform. These limitations highlight the need for more seamless and automated BIM–BI integration frameworks.

Lopes and Boscarioli (2020) conducted a systematic literature review to examine Business Intelligence and analytical tools that support management processes within the construction sector [28]. The study analyzed 1407 articles from multiple databases. A rigorous selection process was implemented to ensure the inclusion of relevant studies only. One of the selection criteria was that the paper should directly address Business Intelligence in the construction, rather than simply referencing it within the text. Out of 1407 studies, only 93 studies were used for further analysis. The results revealed that 35.48% of the reviewed studies employed data warehouses, 37.63% utilized data mining techniques, and 22.58% applied Online Analytical Processing (OLAP) methods. In contrast, dashboard-based solutions were identified in only 4.30% of the analyzed studies [28]. Analysis shows that the adoption of BI dashboards in the construction sector has been slow over the previous years; thus, this gives us an indication that more studies should be done in this field.

Although few studies have explored the application of Business Intelligence within the construction sector, several limitations remain. To address these limitations, the present study proposes a comprehensive framework for integrating BIM data with the analytical capabilities of Power BI to support real-time monitoring, performance evaluation, and decision-making.

3. Methodology

3.1. Research Framework

In this section, the proposed framework for creating an intelligent dashboard that utilizes the analytical capabilities of Business Intelligence with the data-rich environment of Building Information Modeling for better control and management of construction projects

along with its conceptual approach will be explained in detail. The framework consists of three main phases, classified as Inputs, Business Intelligence Integration, and Outputs as illustrated in Figure 1. Each phase consists of subsequent stages that will be explained in the following sections. The first phase in the proposed framework is Inputs, which represents the data sources feeding into the system. During this stage, various data sources are prepared and linked to Power BI (v2022). The input data includes the 3D models from Revit (v2022) software, encompassing both structural and architectural models. This data provides the geometric and the visual details that will be used in the second phase.

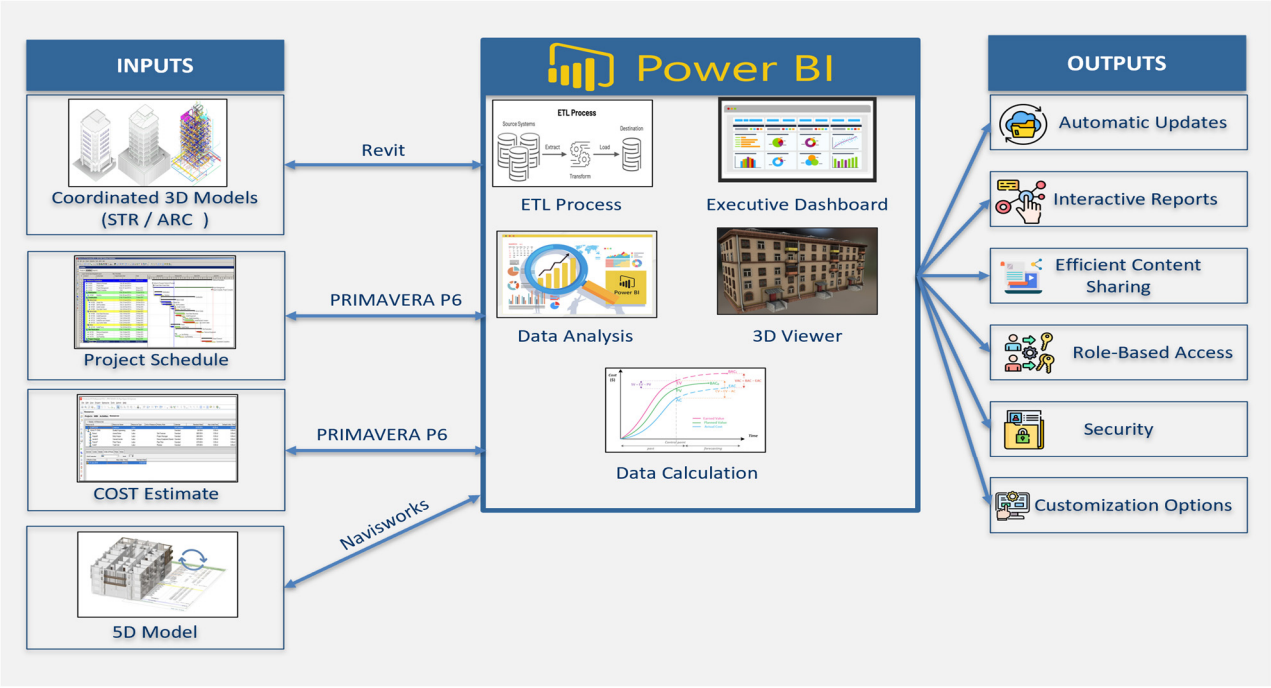


Figure 1. Integrated BIM-BI framework for construction project management.

Project cost estimates and project schedules, developed in Primavera P6 (v2022), represent the second input to the data model. The project schedule includes, but is not limited to, construction activities, activity statuses, and timeline details. The cost estimate, on the other hand, encompasses the actual and budgeted costs of the project. The 5D model developed in Navisworks (v2022) represents the final input to the data model. In general, the Inputs phase consists of five subsequent stages, which will be explained in detail in the following section. The second phase of the proposed framework represents Power BI Integration, where data analysis and integration occur. This phase constitutes the data transformation stage, in which raw data integrated from multiple sources is analyzed to extract meaningful insights. The first stage in this phase is the ETL process, an acronym for Extract, Transform, and Load. The Extract stage represents the initiation of connections between Power BI and the various data sources, along with the determination of the connection type for each data source. The Transform stage encompasses the cleansing and restructuring of the imported data into a format compatible with Power BI's analytical mechanisms. The Load stage involves reintroducing the transformed data into Power BI for analysis and insight generation, as well as establishing the relationships between the different data tables. The last stage in the Power BI Integration phase is visualization and reporting, which includes dashboard design, selection of appropriate visual elements, implementation of interactivity features, and performance optimization.

The final phase represents the Outputs of the integration process. For illustrative purposes, a two-story residential building has been selected for modeling and analysis.

The building is in the Al Nahda district, approximately 15 km from the main road, and comprises two floors. The area on each floor is approximately 206 m^2. Although the methodology has been applied on a two-story residential building, the proposed framework is designed with scalability as a fundamental consideration. This ensures its applicability across a broad spectrum of construction projects regardless of scale or complexity. The major phases of the research framework are illustrated in Figure 1.

3.2. Design Models

The first step in the Inputs phase is generating the design models that will be used in the subsequent analysis conducted in the second phase of the framework. Architectural and structural models were developed using Revit, based on 2D design drawings prepared in AutoCAD (v2022). Revit is widely recognized as one of the primary authoring tools developed to realize the concept of Building Information Modeling, enabling the construction of 3D models derived solely from 2D plans. This technology has demonstrated significant savings in both time and cost compared to conventional methods. The process of generating detailed models begins with exporting grid lines and elevations from the AutoCAD plans. Subsequently, the structural model was developed in a separate file using the structural layout template as the base input. Construction of the structural model commenced with the foundation, followed by neck columns and grade beams, and progressed through to the roof slab. The architectural model was then developed following the same approach. Upon completion of the first phase, two detailed models are produced: the structural model and the architectural model.

The design review phase commenced with an inspection of the models developed in the preceding stage to identify any internal conflicts, followed by the combination of the two models into a single file to detect any preliminary errors or clashes. The identified clashes are subsequently communicated to the designers to make the required adjustments to the models. This process yields considerable savings in both time and cost, compared with if these clashes had just been discovered during the construction work. Upon receiving the revised designs, the updated changes are incorporated into the model and subjected to further testing to verify the absence of any remaining errors.

3.3. 3D Modeling Phase

Revit is a highly effective software platform utilized for 3D modeling, designed specifically for use by architects, engineers, and construction professionals throughout the project lifecycle. It has experienced a high rate of adoption within the construction sector in recent years, owing to its powerful capabilities and streamlined processes. Revit utilizes a parametric modeling system that enables users to create and manipulate building elements such as walls, floors, roofs, doors, and windows, while maintaining their relationships. This parametric method plays an important role in design repetition and rework. Revit was employed in this study to produce coordinated, information-rich 3D models for both the structural and architectural components of the building. The structural modeling commenced with the assembly of all structural elements, including but not limited to foundations, columns, slabs, structural framings, and stairs. Detailed reinforcement was prepared for each element. This step presented considerable challenges, as Revit's reinforcement functionality requires each element to be detailed manually, which consumes a lot of time and effort. It is worth noting that enhanced automation of reinforcement detailing in future versions of the software would substantially improve its practicality for this purpose. Upon completion of the structural model, the same methodology was applied to the architectural model. All architectural components were assembled, including but not limited to floors, windows, doors, walls, and plastering, along with their respective properties

and specifications. The resulting 3D architectural model proved instrumental in resolving several design discrepancies, such as inconsistencies between floor plans and elevations. The 3D architectural and structural models are illustrated in Figures 2 and 3, respectively.

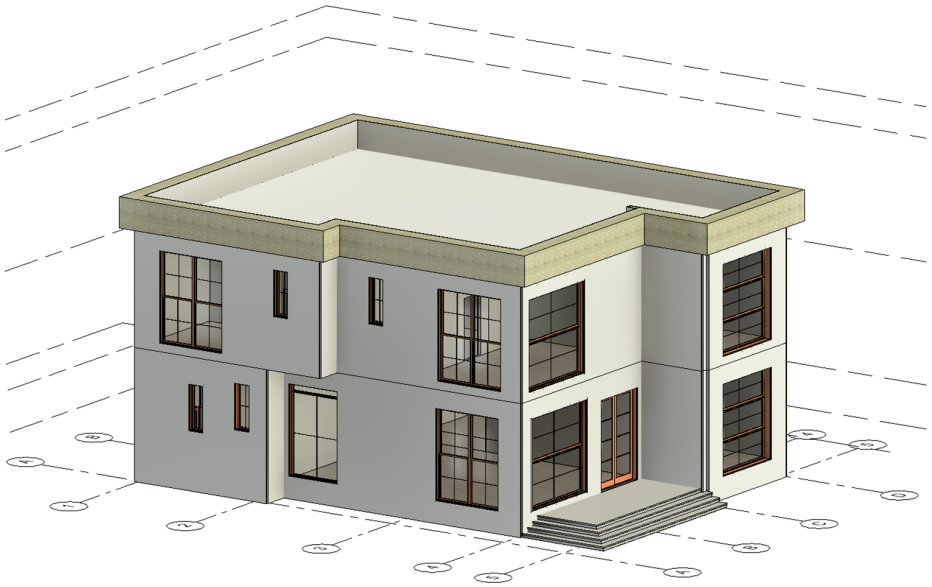


Figure 2. Architectural 3D model of case study.

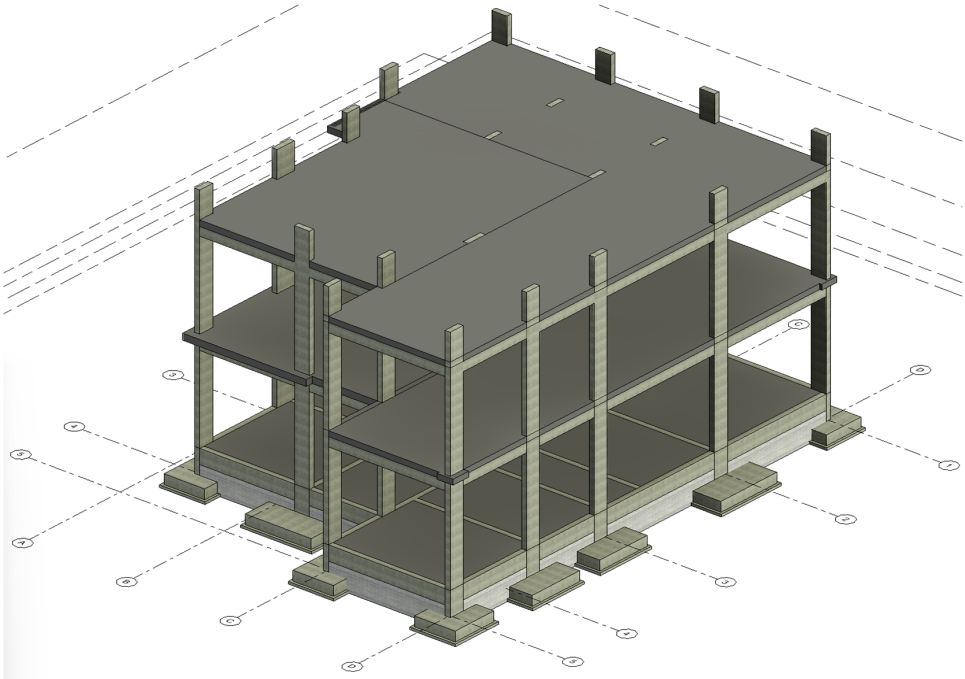


Figure 3. Structural 3D model of case study.

Following the design review, the structural and architectural models were combined into a single coordinated 3D model. Clash detection tools were employed to identify interferences between systems, a process that substantially reduces design errors during construction by simulating the real-world installation of building systems. All identified clashes between the two models were resolved prior to their consolidation into a unified model. Upon confirmation that the integrated model was free of conflicts, the subsequent step of preparing quantity take-offs for both the structural and architectural models was initiated. The quantity take-off was conducted using Revit software, owing to its accuracy

and efficiency. The advanced capabilities of Revit enabled the automated extraction of quantities for all elements across both models within a short timeframe. These quantities serve as the basis for the subsequent stage, which involves the planning and estimation of the project schedule and cost, whereas project schedule and activity duration are calculated based on project quantity take-off sheets. A sample of the quantity take-off for the foundation is illustrated in Figure 4.

Structural Foundation Schedule

Type	Length	Width	Foundation Thickness	Volume	Area
F 1					
F 1	1.80	1.60	0.40	1.15 m³	3 m²
F 1	1.80	1.60	0.40	1.15 m³	3 m²
F 1	1.80	1.60	0.40	1.15 m³	3 m²
F 1	1.80	1.60	0.40	1.15 m³	3 m²
F 1	1.80	1.60	0.40	1.15 m³	3 m²
F 1	1.80	1.60	0.40	1.15 m³	3 m²
6				6.91 m³	17 m²
F 2					
F 2	2.20	2.00	0.40	1.76 m³	4 m²
F 2	2.20	2.00	0.40	1.76 m³	4 m²
F 2	2.20	2.00	0.40	1.76 m³	4 m²
F 2	2.20	2.00	0.40	1.76 m³	4 m²
F 2	2.20	2.00	0.40	1.76 m³	4 m²
F 2	2.20	2.00	0.40	1.76 m³	4 m²
6				10.56 m³	26 m²
F 3					
F 3	2.60	2.40	0.40	2.50 m³	6 m²
F 3	2.60	2.40	0.40	2.50 m³	6 m²
F 3	2.60	2.40	0.40	2.50 m³	6 m²
F 3	2.60	2.40	0.40	2.50 m³	6 m²
F 3	2.60	2.40	0.40	2.50 m³	6 m²
5				12.48 m³	31 m²
F 4					
F 4	3.00	2.60	0.40	3.12 m³	8 m²
F 4	3.00	2.60	0.40	3.12 m³	8 m²
2				6.24 m³	16 m²
F 5					
F 5	4.00	3.60	0.50	7.20 m³	14 m²
1				7.20 m³	14 m²
PC Footings					
PC Footings	19.40	15.40	0.10	12.31 m³	123 m²
1				12.31 m³	123 m²
Grand total: 21				55.71 m³	228 m²

Footing Rebar Schedule

Partition	Type	Bar Diameter	Bar Length	Quantity	Shape Image	A	B	C	Length of each bar	Total Bar Length	steel weight (Ton)
F1											
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
F1	Ø12	12 mm	2262 mm	11		300 mm	1699 mm	300 mm	2265 mm	24.88 m	0.022114
F1	Ø12	12 mm	2062 mm	13		300 mm	1499 mm	300 mm	2065 mm	26.80 m	0.023823
											0.275621
F2											
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
F2	Ø12	12 mm	2462 mm	15		300 mm	1900 mm	300 mm	2466 mm	36.93 m	0.03283
F2	Ø12	12 mm	2662 mm	14		300 mm	2100 mm	300 mm	2666 mm	37.27 m	0.03313
											0.395764
F3											
F3	Ø12	12 mm	2862 mm	18		300 mm	2300 mm	300 mm	2866 mm	51.51 m	0.045789
F3	Ø12	12 mm	3061 mm	17		300 mm	2499 mm	300 mm	3065 mm	52.04 m	0.046262
F3	Ø12	12 mm	2862 mm	18		300 mm	2300 mm	300 mm	2866 mm	51.51 m	0.045789
F3	Ø12	12 mm	3061 mm	17		300 mm	2499 mm	300 mm	3065 mm	52.04 m	0.046262
F3	Ø12	12 mm	2862 mm	18		300 mm	2300 mm	300 mm	2866 mm	51.51 m	0.045789
F3	Ø12	12 mm	3061 mm	17		300 mm	2499 mm	300 mm	3065 mm	52.04 m	0.046262
F3	Ø12	12 mm	2862 mm	18		300 mm	2300 mm	300 mm	2866 mm	51.51 m	0.045789
F3	Ø12	12 mm	3061 mm	17		300 mm	2499 mm	300 mm	3065 mm	52.04 m	0.046262
F3	Ø12	12 mm	2862 mm	18		300 mm	2300 mm	300 mm	2866 mm	51.51 m	0.045789
F3	Ø12	12 mm	3061 mm	17		300 mm	2499 mm	300 mm	3065 mm	52.04 m	0.046262
											0.460257
F4											
F4	Ø16	16 mm	3053 mm	16		300 mm	2500 mm	300 mm	3058 mm	48.84 m	0.07718
F4	Ø16	16 mm	3453 mm	14		300 mm	2900 mm	300 mm	3458 mm	48.34 m	0.076382
F4	Ø16	16 mm	3053 mm	16		300 mm	2500 mm	300 mm	3058 mm	48.84 m	0.07718
F4	Ø16	16 mm	3453 mm	14		300 mm	2900 mm	300 mm	3458 mm	48.34 m	0.076382
											0.307126
F5											
F5	Ø16	16 mm	4252 mm	21		400 mm	3499 mm	400 mm	4257 mm	89.28 m	0.141089
F5	Ø16	16 mm	4652 mm	19		400 mm	3900 mm	400 mm	4658 mm	88.39 m	0.139682
											0.28077
											1.719538

Figure 4. Foundation quantities.

3.4. Planning Phase

Primavera is a powerful project management tool developed by Oracle Corporation, widely utilized across industries such as construction and manufacturing. It offers plenty of features and functions for project planning, scheduling, and control. Primavera P6 supports the creation of Work Breakdown Structures (WBS) along with the identification and sequencing of project activities. It schedules using the Critical Path Method (CPM),

as well as providing visual representation of project activities through Gantt charts. Furthermore, it facilitates resource allocation and management across various resource types, including labor, equipment, and materials. The key function of Primavera lies in its ability to monitor actual progress against the planned schedule throughout the project lifecycle. This process can be further enhanced through the integration of Power BI, which enables streamlined progress in a 3D visually appealing dashboard. In the present study, Primavera was employed to develop the project schedule, with the primary objectives of estimating the time required to complete the project and determining the total project cost. The initial phase of project planning involved decomposing the project into hierarchical levels through the development of a Work Breakdown Structure (WBS).

The WBS represents a systematic decomposition of the overall project into smaller, more manageable components known as work packages. Each work package comprises a set of related activities that share common characteristics. This structured approach facilitates a clear visualization of the project scope and deliverables. The project was divided into three primary categories: site preparation, earthwork, and floor work. Each floor work package consists of concrete work and finishing work, with smaller work packages containing lists of related activities nested beneath each category. Following the decomposition of the project into work packages and activities, durations were assigned to each activity using the following formula: Duration = Quantity/Productivity. Quantities for each activity were extracted from the corresponding Revit model developed in the preceding stage. Productivity rates were determined through a combination of experience and consultation with professional engineers. Activity relationships and dependencies were subsequently assigned based on construction logic and the intended construction methodology. A sample of the project schedule is illustrated in Figure 5. Resources were then assigned to each activity to calculate the associated costs. The cost of each activity was calculated based on the quantity and unit price of the respective resources.

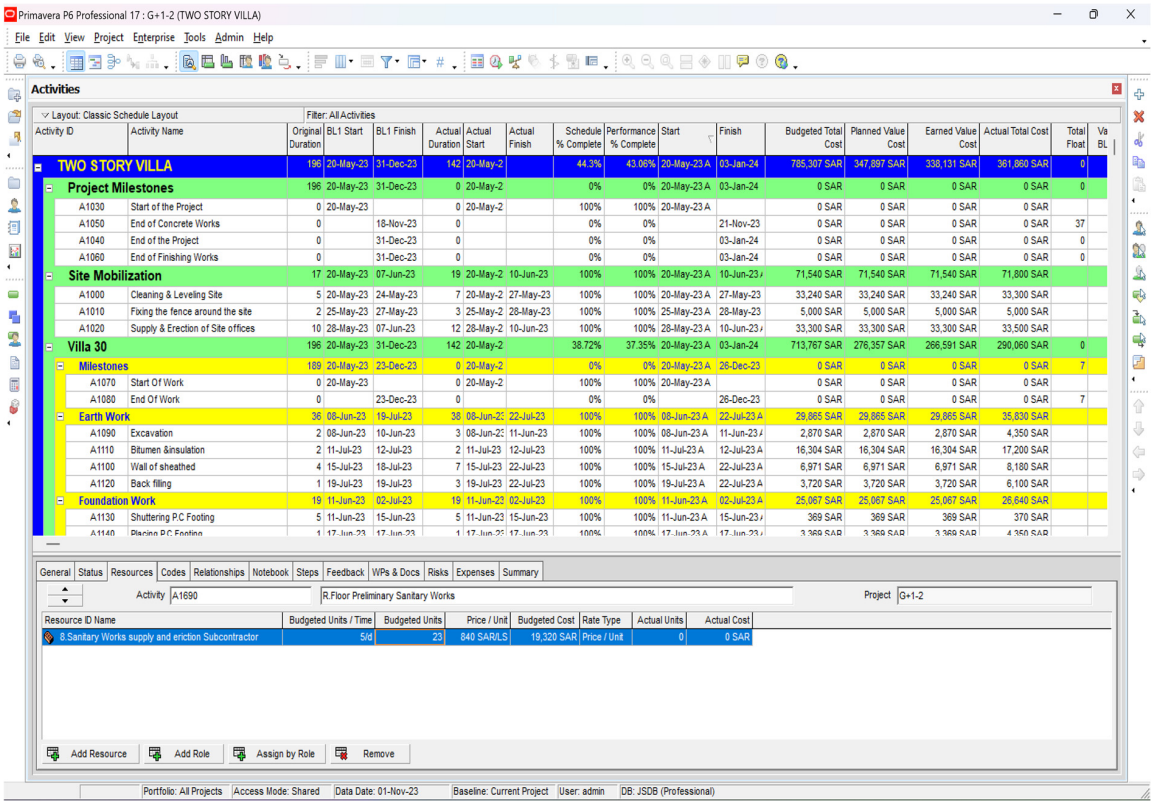


Figure 5. Sample of project schedule.

3.5. 5D Modeling Phase

5D modeling represents an advanced approach of modeling and visualization that incorporates the dimensions of cost and time into the standard 3D modeling process, offering a more detailed and comprehensive representation of construction projects. This methodology has been widely adopted in the construction industry to improve collaboration, reduce errors, increase efficiency, and minimize delays and cost overruns. It provides the capability to simulate construction processes, evaluate alternative scenarios, identify potential issues at an early stage, and support decision-making prior to the commencement of construction. The simulation was initiated by importing the project schedule and cost estimate from Primavera. As Navisworks doesn't support the Primavera schedule format, the schedule was first converted to *.xls format before being imported into Navisworks. To ensure accurate mapping between each activity and its corresponding element in the model, adjustments were made to the 3D model in Revit prior to its import into Navisworks. Within Primavera, each activity is assigned a unique identification code. This code was extracted from each activity and assigned to the corresponding element in the 3D model.

Following the assignment of identification codes to each element, both the structural and architectural models were imported into Navisworks to commence the 5D simulation. The simulation was conducted to mimic the construction activities of the project, verify the logic and sequence of the construction elements, identify potential issues during the construction process, track project cost and schedule in a visualized mode, and make decisions prior to the actual commencement of construction. Upon ensuring that all elements were properly coordinated, the study proceeded to the second phase of the proposed framework. A snapshot of the 5D simulation is illustrated in Figure 6.

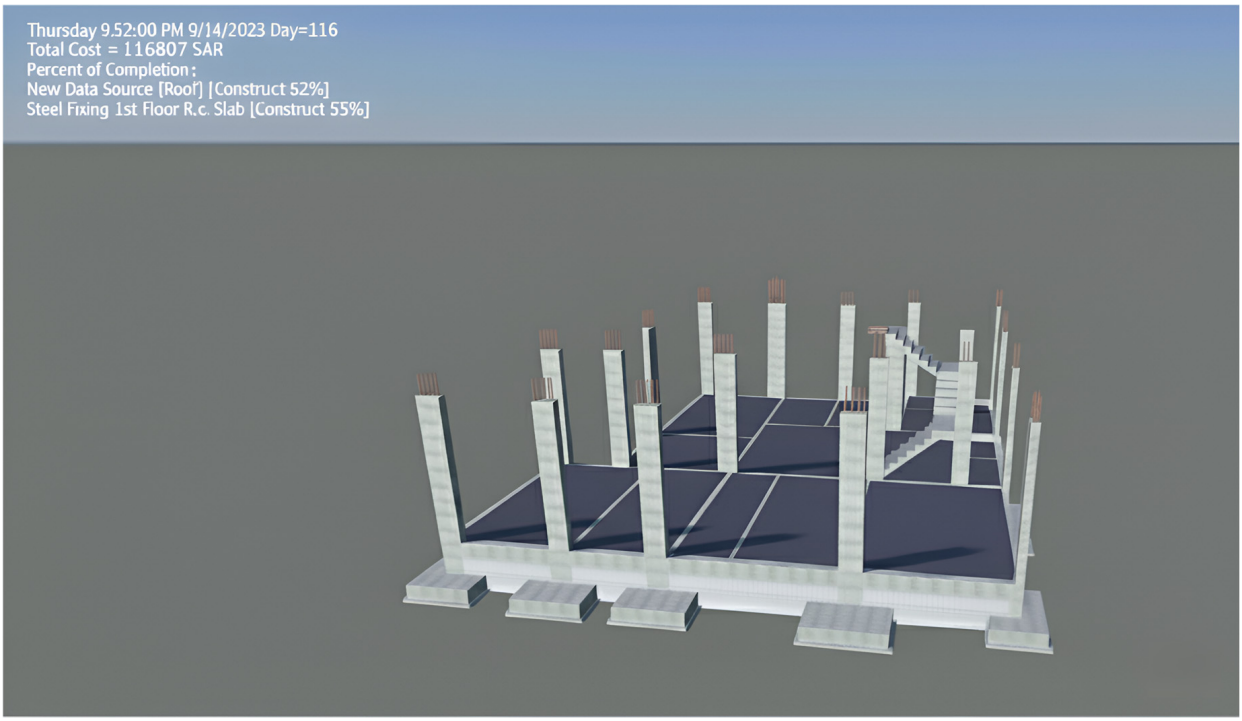


Figure 6. 5D simulation of case study.

3.6. Business Intelligence Integration

Cost overruns, schedule delays, and quality control issues represent persistent challenges within the contemporary construction sector. In this context, leveraging data analytics technologies such as Power BI presents a significant opportunity to address these challenges. Power BI offers real-time updated dashboards, interactive visualizations, and

integration of data from multiple sources which gives stakeholders, especially project managers, the ability to make reliable decisions rapidly. The application of Power BI for project monitoring and control follows a structured series of steps. The first step is identifying key stakeholders involved. Stakeholder experience is a crucial component of data analysis and visualization, referring to the distinct needs, interests, and expectations of those who interact with the visualizations and insights presented within a dashboard report. These stakeholders may include, but are not limited to, managers, executives, engineers, owners, and any party with a special interest in the data insights. To accomplish this step, consultations were conducted with several construction project managers to gather their insights, requirements, goals, and scope of work with respect to Power BI. The feedback indicated that project managers are primarily concerned with cost tracking, schedule monitoring, and the highlighting of key performance indicators. These requirements were given considerable emphasis throughout the dashboard design process to help project managers monitor project efficiency and make well-informed decisions.

The second step involves the application of ETL methodology to prepare the data for analysis. Power BI supports data extraction from a wide range of sources, including Excel spreadsheets, cloud-based data sources, third-party data services, and other data sources such as web connectors, SurveyMonkey, and custom API connectors. During this study, the previously developed models were connected to the Autodesk Platform Services (APS) [v2022], after which a custom API connector was utilized to extract the model data into Power BI Desktop. Power BI offers two primary connectivity modes for linking data sources to Power BI for analysis and report generation: Import Mode and Direct Query.

Import Mode copies data from the source into Power BI's memory, enabling highly responsive interactive analysis and fast performance, whereas in Direct Query, rather than transferring data, Power BI maintains a live connection to the data source and queries it in real time. This is particularly important when dealing with large volumes of data or when instant information is required, for example when obtaining data from sensors installed at various locations on the construction site that necessitate immediate action in response to changes in data. However, depending on the source system's availability and latency, there could be performance trade-offs. In the present study, Import Mode was adopted given the manageable size of the dataset and the processing limitations of the hardware used for analysis and dashboard development. Once the data has been extracted and connected to Power BI, transforming steps commence, where raw data is transformed and reshaped to meet the analytical requirements. This stage is carried out primarily within the Power Query Editor, which provides a comprehensive toolset for cleaning, transforming, filtering, and manipulating data through an intuitive interface that operates using the M coding language. The data transformation stage involves two processes, cleaning and transformation of data. The cleaning process involves removing nulls or data errors, replacing values, and the inclusion or exclusion of specific rows and columns. The transformation process involves modifying data types such as time, integer, date, text, etc. It also involves pivoting and unpivoting data, merging and appending queries, and splitting or combining columns. The transformation phase is very critical and should be closely monitored to produce a reliable dashboard. Power Query preserves a record of every step performed during the transformation process, enabling steps to be undone, modified, or validated at any stage.

The load phase represents the final step of the Power BI ETL process, in which the cleaned and transformed data is loaded into the Power BI data model. At this stage, the transformed data becomes accessible for analysis, enabling the creation of interactive reports, visualizations, and dashboards. Once the data is loaded into the Power BI data model, the subsequent step involves defining the relationships between the various tables,

since our data has been gathered from different sources. There are three types of relationships that can be used in the model: One-to-Many, Many-to-One, and Many-to-Many. Properly defined relationships are essential to ensuring that reports are interactive. Figure 7 illustrates the different relationships between tables within the model view of Power BI.

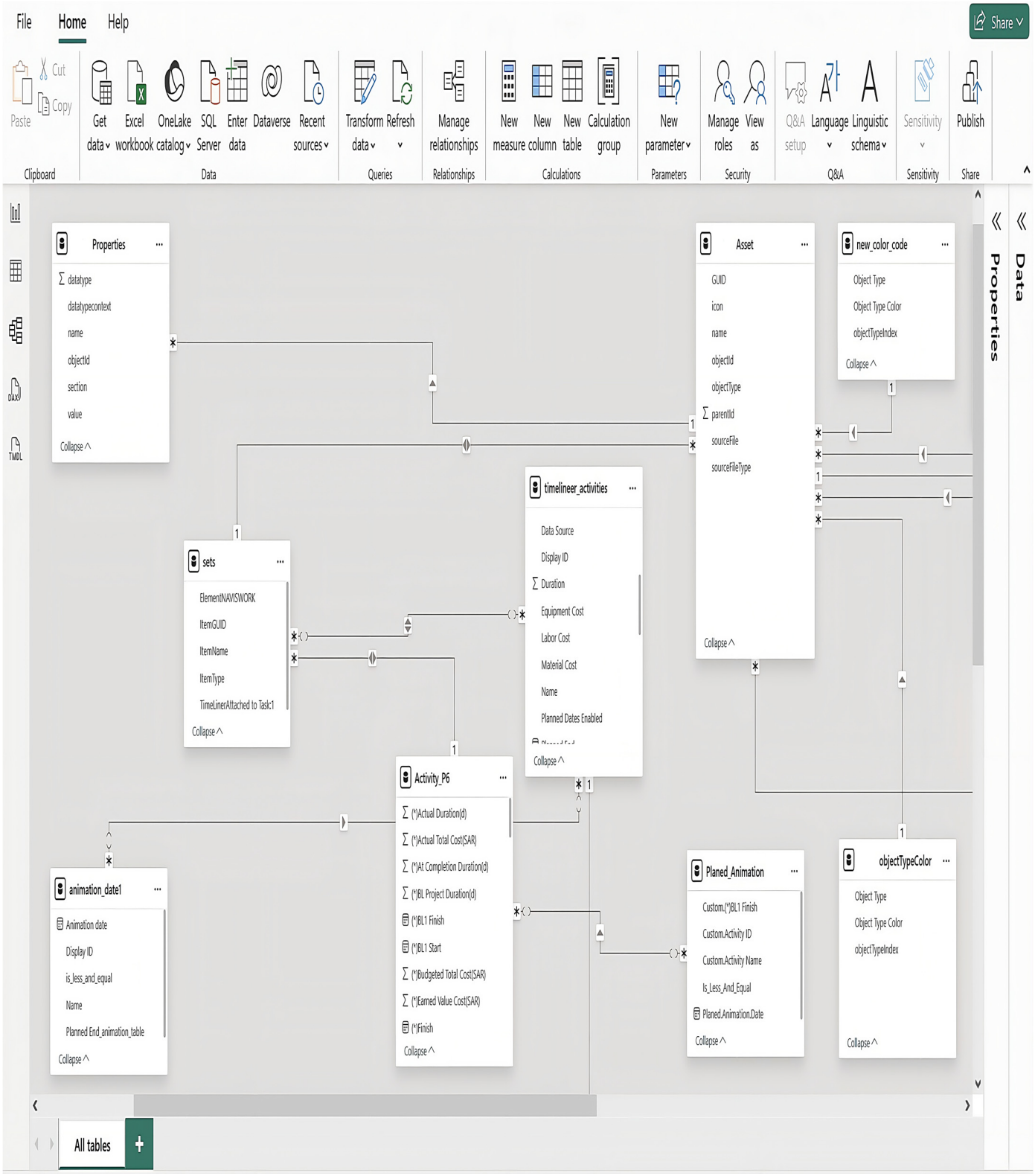


Figure 7. Power BI model view.

The Asset table serves as the central reference point within the Power BI data model, playing a pivotal role in structuring and organizing project information. This table houses a comprehensive list of all object IDs corresponding to every element present in both the structural and architectural models. In addition to the object ID, the Asset table contains other key attributes essential for identifying project components, including source file, name, object type, and GUID. The source file attribute specifies the origin of the data for each object, enabling users to distinguish between different object types and their respective sources. The Asset table contains another important field, name, which identifies the type of elements present in the imported models, including doors, windows, floors, footings, etc. Each element in the imported models possesses a unique identifier recorded in the object type field, such as "D1" for doors, "W1" for windows and "F1" for footings. This naming system functions as an effective tool for the rapid recognition and filtering of components during the analysis and visualization of data within the Power BI dashboard.

During the analysis phase in Power BI, the loaded data was transformed into actionable insights through the construction of robust semantic models, crafting DAX calculations, and conducting exploratory investigations. This phase begins after the data has been extracted, transformed, and loaded, and ends up with a validated set of metrics and preliminary insights that are ready for visualization and storytelling. This step bridges the data preparation and reporting phases by enabling users to uncover patterns, trends, and insights through exploring relationships within the data, creating calculations and metrics, and designing interactive visualizations. The analysis phase was conducted using Power BI Desktop to convert cleaned data into Business Intelligence insights. The foundation of the analysis phase lies in the use of Data Analysis Expressions (DAX), the formula language that powers all custom computations in Power BI. DAX enables the creation of dynamic measures, calculated columns, and tables that compute key metrics at query time based on filter context. Furthermore, DAX enables users to build analytical custom models. These models surpass static reporting and support deeper exploration of BIM datasets. Project stakeholders can use them to obtain customized metrics, perform cross dataset calculations, and conduct scenario-based evaluations. DAX was employed to calculate several metrics, including the cumulative cost for planned, actual, and earned value costs over the project duration. These calculated measures were subsequently utilized during the reporting phase to demonstrate key performance indicators of the project, including addressing critical project status questions such as whether the project is behind or ahead of schedule, and whether costs are below or exceeding the budget.

A range of functions including sum, subtract, divide, calculate, and filter were employed to construct both simple aggregations and advanced calculations in accordance with the requirements of the interactive dashboard. By the conclusion of this phase, a comprehensive collection of validated DAX measures had been developed to capture the project's key performance indicators and preliminary insights. The results from this stage serve as the basis for the visualization and reporting phase, wherein decision-makers can see the analysis brought to life through the application of design best practices and data narrative. Figure 8 illustrates a sample DAX formula used to calculate the cumulative budgeted total cost of the project.

Figure 8 illustrates the process and the code developed for calculating the cumulative budgeted total cost of the project in Power BI. The CALCULATE function modifies the filter context by determining which rows are included in the summation, aggregating all rows up to and including the current month. Without this function, the calculation would only sum the cost of the current row or month. The SUM function aggregates the values of the budgeted total cost from the Activity_P6 table. SUMMARIZE creates a summary table by categorizing dates into months and corresponding month numbers, whereby January is

assigned month number one and December is assigned month number twelve. This step is critical for grouping activity dates with their corresponding months. The ALLSELECTED function introduces interactivity to the dashboard by respecting user selections made within the slicers, ensuring that the calculation of the budgeted total cost operates across all selected months. For example, if the user selects only the months from January to March, the table will include only month numbers one through three in the calculation of the budgeted total cost. The ISONORAFTER function is used to evaluate whether each row in the month table corresponds to the current month or any preceding month. For example, we assume the current month is March (month number = 3), and the current date is the 30th. Upon execution, the function evaluates each month sequentially. January returns a true value, as it precedes March. February similarly returns a true value, as it also precedes the current month. March is then evaluated, with the function additionally checking the day value; a value of true is returned only if the date falls on or before the 30th of March. April returns a value of false, as it succeeds March. The FILTER function is used to filter the month table, retaining only those rows for which the ISONORAFTER value is true. The final step in the formula is the CALCULATE function, which computes the summation of the budgeted total cost across the selected months. In conclusion, this formula constructs a running total of budgeted costs on a month-by-month basis, while respecting slicer selections and user interactions within the dashboard.

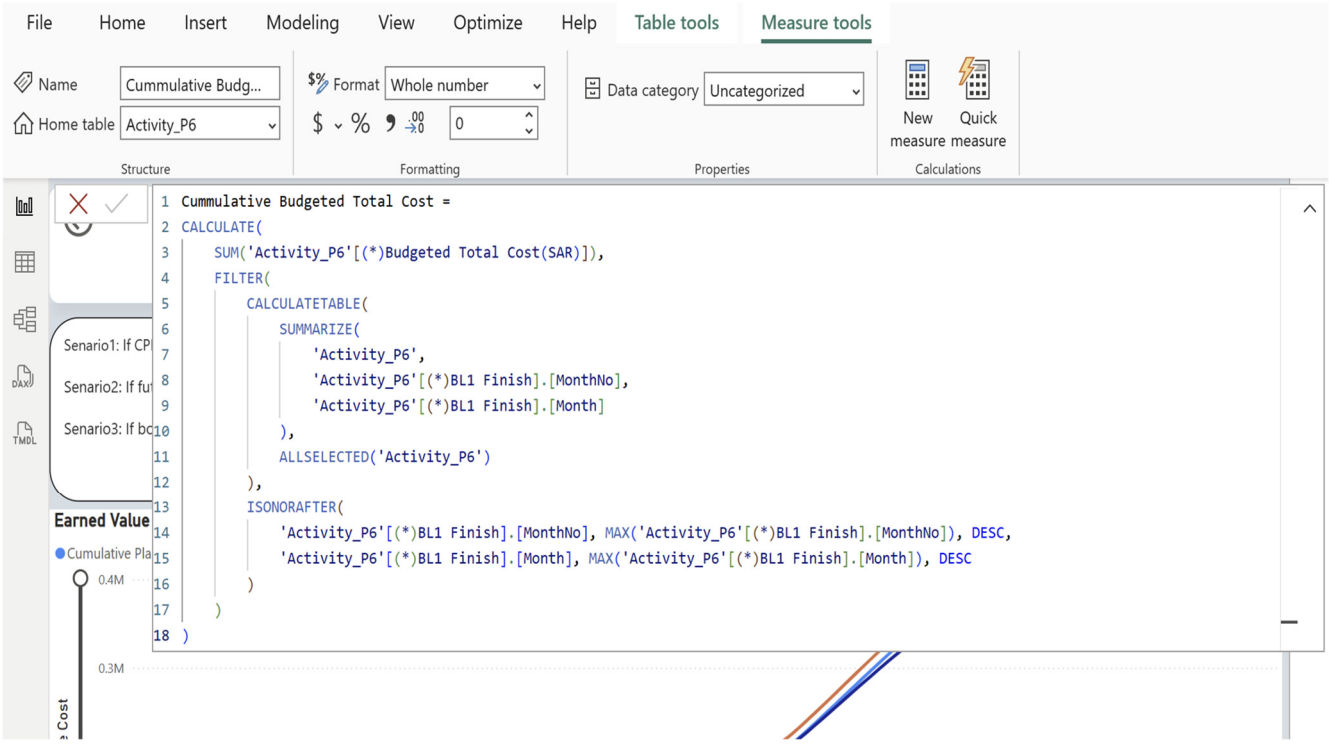


Figure 8. Data Analysis Expression (DAX) for cumulative budgeted total cost.

The visualization and reporting phase encompasses several key design principles, including layout, visual selection, interactivity, performance optimization, and regulated distribution. This is achieved through the application of best practices for audience-centric storytelling and appropriate chart selection. Upon completion of the analytical phase, Power BI Desktop was utilized to convert the analyzed model into an interactive visual dashboard that facilitates data-driven decision-making. Effective report design begins with a thorough understanding of end users and their objectives, a requirement addressed in the preceding stages through stakeholder consultation. When developing the dash-

board, considerable effort was directed towards three principal elements: designing and formatting interactive visuals, organizing insights into a cohesive dashboard layout, and incorporating interactivity to support user exploration. Power BI offers a comprehensive library of built-in visuals, including standard charts such as column and bar charts, line charts, and scatter plots; geographic maps such as standard maps and Azure Maps; matrix and table visuals; KPIs such as gauge charts; and more advanced custom visuals available through the Marketplace.

Selecting the appropriate visual for the data is essential to the development of an intelligent dashboard. For example, line charts are suited to tracking trends over time, tables and matrices to display detailed data, bar and column charts to compare categories, and card visuals to highlight key performance indicators. To enhance interactivity and user experience, slicers and filters were incorporated into the dashboard. The dashboard was further augmented with a smart search function, enabling users to retrieve information by entering a search term or as few as the first three characters, whereupon Power BI's intelligence capability scans all imported data and provides results promptly. Furthermore, a cross-highlighting feature was integrated into the dashboard, whereby interacting with a data element automatically highlights all related information while rendering other data transparent, thereby enhancing the user experience by providing intelligence to the report. Upon completion of the dashboard customization process, the report was published to Power BI Service.

Power BI Service is the cloud-based platform of Microsoft Power BI, accessible through any web browser, and designed for publishing, sharing, and collaborating on interactive dashboards and reports. Once published, limited customization of the dashboard remains possible; however, its primary function is to share reports with different stakeholders based on pre-defined roles and to facilitate real-time collaboration. Within Power BI Service, a workspace was created to serve as the central container for datasets, reports, and dashboards, as well as a collaboration platform for relevant stakeholders. A significant feature of this workspace is its integration with Microsoft Teams, which enables real-time discussions among stakeholders. To manage and organize the workspace effectively, four roles are assigned to the respective stakeholders: Admin, Member, Contributor, and Viewer. The Admin role represents the highest level of authority, granting full control over the workspace, including the ability to add or remove users; create, edit, or delete content; and delete the workspace entirely. This role is typically assigned to the most senior entity within the project.

The Member role permits the editing, deletion, and publishing of content within the workspace, making it particularly suitable for report developers and analysts given their capacity to contribute substantively to the workspace. The Contributor role is limited to editing content within the workspace. This role is ideal for junior analysts due to their limited scope of responsibility. The Viewer role carries minimal responsibilities, permitting holders only to view content within the workspace. This role is ideally assigned to end users who require access solely to the insights presented within reports or dashboards. Understanding the roles in Power BI is critical for effective collaboration and maintaining the security of the content within the workspace. Power BI Service provides the option to share reports publicly via the web, whereby any individual in possession of the link may access the dashboard. However, this method does not permit tracking of viewers or control over redistribution. That is why it's recommended when sharing reports or dashboards to set roles for those who can access and interact with the content, especially when it contains sensitive information. The preparation process for Power BI to control and manage projects is illustrated in Figure 9. To streamline the methodology used in the study, a process chart was created as illustrated in Figure 10.



Figure 9. Power BI preparation process.

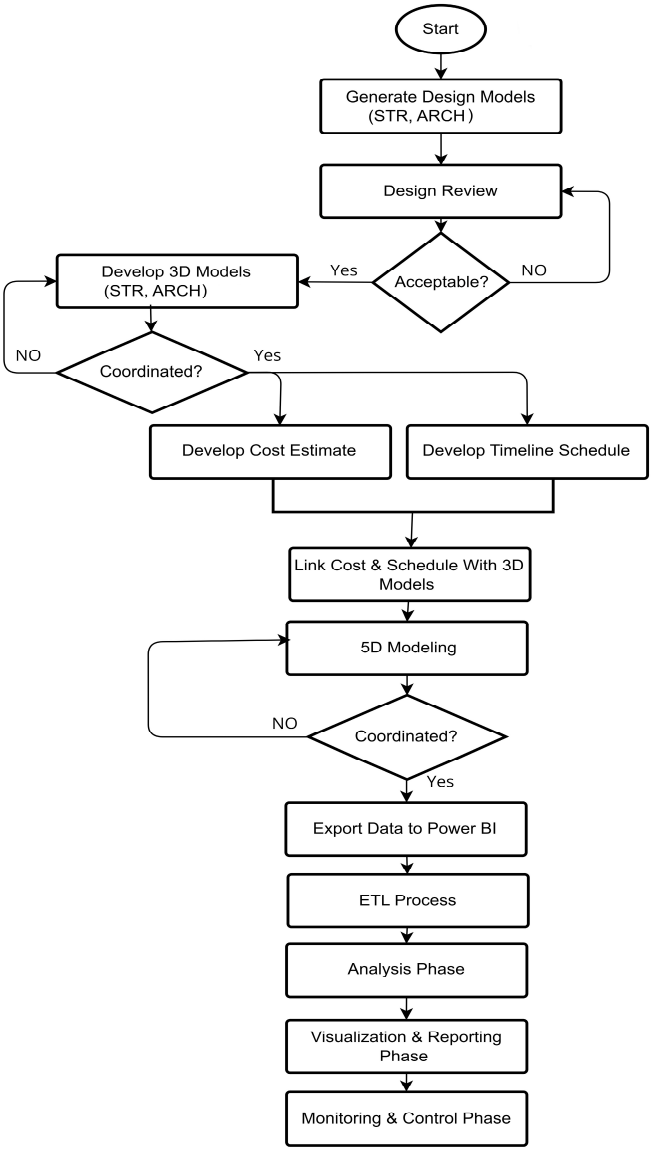


Figure 10. Methodology process chart.

4. Results and Discussion

Benefits of the Customized Dashboard

The customized dashboard provides accurate, real-time, and cross-referenced visualizations of project data, ensuring that the insights displayed are based on the most current data available. By integrating the Business Intelligence capabilities of Power BI with BIM and automating the updates of data, the dashboard will provide stakeholders with the abil-

ity to enhance decision-making and enhance overall project management practices. During this study, the dashboard was subjected to a series of detailed analyses and evaluations to assess its capabilities. This exploration has resulted in the identification of four major advantages of using a Power BI dashboard that utilizes BIM.

The foremost advantage is the ability to provide real-time data updates and generate interactive reports. Construction projects are characterized by their fast-paced development, thus having access to the most recent data is critical to managing these projects efficiently. Through Power BI, stakeholders can visualize data as it is generated, which enables them to have immediate insights into project metrics. In addition to improving decision-making, this instance access enables project stakeholders to respond promptly to emerging issues, thereby keeping the project on track within the specified timeline and budget. Figure 11 presents a summary dashboard of project progress that updates in real time.

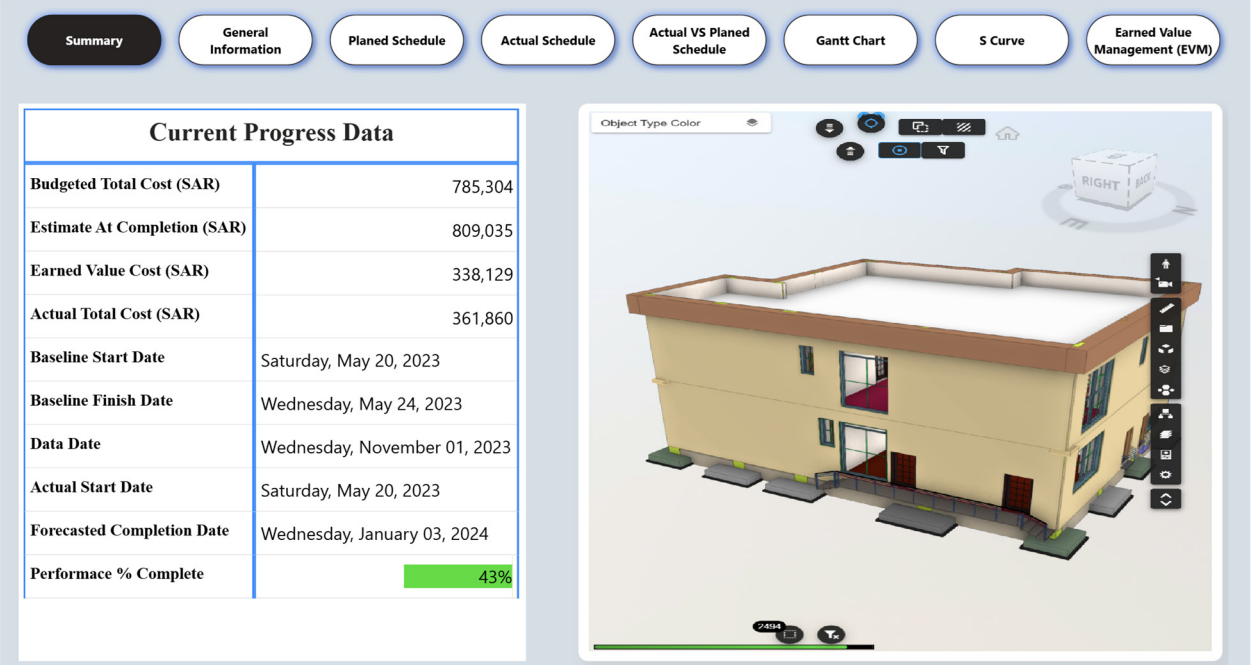


Figure 11. Power BI summary dashboard.

This dashboard provides a concise summary of the project's current progress, highlighting key information related to project cost, schedule, and overall performance. The cost information encompasses the budgeted total cost of the project, the forecasted estimate at completion based on current performance trends, the earned value cost, and the actual total cost incurred to date. Collectively, these metrics enable stakeholders to gain immediate insight into the project's financial health, including whether it is on budget, over budget, or under budget. Based on this information, project managers can take timely corrective actions and well-informed decisions in real time to support the achievement of project goals and requirements.

Furthermore, the dashboard presents information related to the project schedule, including the baseline start and finish dates, the actual start date, the current date, and the forecasted completion date. These details provide stakeholders with quick insights and accurate understanding of the project's timeline status. Additionally, the dashboard presents performance status information, reflected as a percentage indicating the overall performance completion of the project. One of the most valuable features of this dashboard is the automatic update, ensuring that the latest project progress information is consistently reflected. For instance, the project data, including invoices, material submittals, and

the actual start and finish dates of activities, are first updated in Primavera P6. Since a connection was established in the previous stages between Primavera P6 and Power BI, every time the project schedule is updated, the dashboard data updates correspondingly. In addition, automatic refresh intervals can be configured, for example, every 30 min, even if no new schedule updates have been made. Moreover, the dashboard supports interactive reporting, as illustrated in Figure 12. When a user interacts with any visual element, the report is automatically filtered to display only the information relevant to the selected component, thereby providing users with an interactive and intelligent experience when exploring the reports.

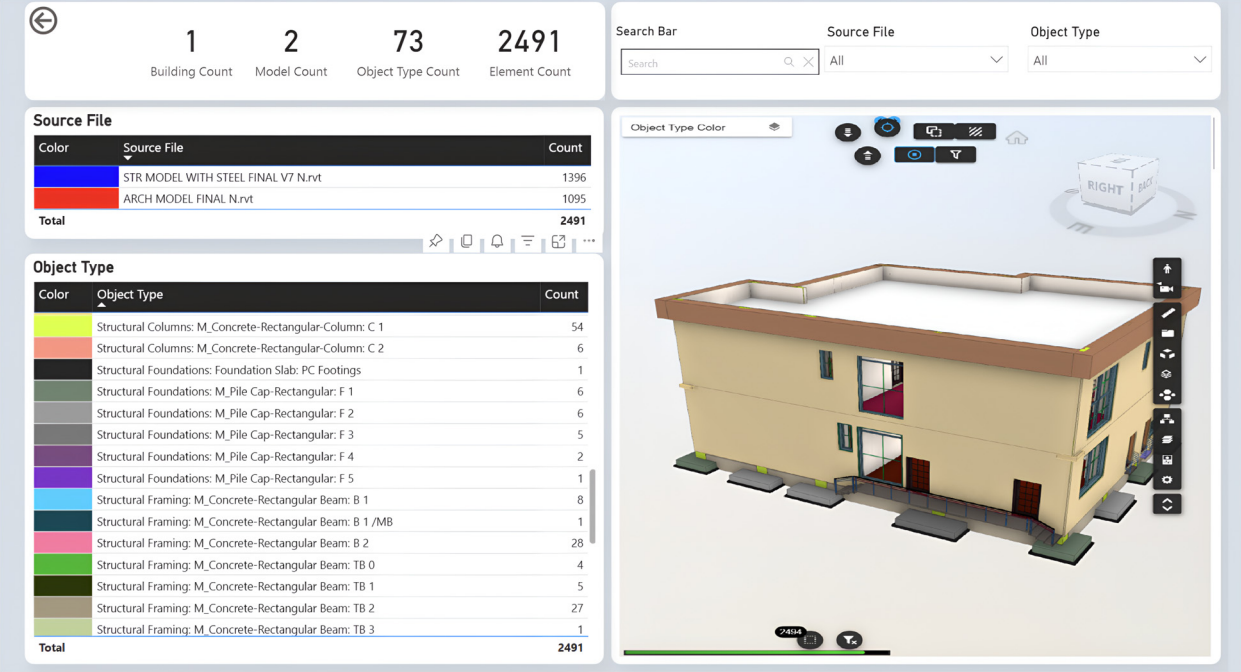


Figure 12. Power BI general information dashboard.

Figure 12 presents general information about the project, including the number of buildings, models, objects, and elements. The layout is organized into five sections. The first section provides an overview of the project quantities. The second section provides a detailed account of the number of models integrated into Power BI for analysis; in this case, two models: the structural and the architectural. Each model is assigned a distinct color to clearly differentiate between them. Along with that, the dashboard displays the number of elements associated with each model. Since interactivity is a common characteristic of our dashboard, when a stakeholder clicks on any model type, the 3D viewer automatically filters to display only the elements related to the chosen model. The third section contains a table listing all objects within the project, with each object assigned a unique color to facilitate clear visual distinction. The table additionally displays the number of elements associated with each object. When a user clicks on any object, the 3D viewer automatically filters the model to display only the selected object and highlights it using the corresponding color system shown in the table. This color system is highly useful, as it enhances clarity and visual navigation. Stakeholders may also select multiple objects simultaneously, enabling them to efficiently analyze, compare, and identify specific components within the 3D environment.

The fourth section focuses on selection and search functionality. Stakeholders can select objects directly from the table or utilize advanced slicers for more efficient filtering and inspection. Three selection functions are embedded within the dashboard. The first

enables users to select single or multiple project objects using object type slicers. For more targeted selection, whereby a user wishes to filter elements associated with a specific model, the file source slicer may be employed. This enables stakeholders to focus on elements from a particular model without the need to navigate through all objects in the project. The most innovative feature within this section is the intelligent search function, which incorporates artificial intelligence capabilities. When a user enters the first three characters of the element being searched, the AI generates relevant suggestions based on the input, providing the option to select from the suggested results or continue entering the full element name. This functionality yields considerable savings in time and effort when retrieving information required to support well-informed decisions. Furthermore, any element selected through the smart search function is automatically highlighted within the 3D viewer.

Finally, the 3D viewer is fully interactive and equipped with multiple advanced features, including a walk-through mode. This function enables stakeholders to navigate through the 3D model as though physically present on the construction site, allowing them to closely examine element materials, finishes, object placement, room layouts, and the interaction between various building components. This capability facilitates the early identification of potential issues and supports the timely implementation of corrective actions before problems escalate. Furthermore, while navigating through the model, stakeholders have also the ability to measure heights, elevations, lengths, and the areas of various objects. This is extremely valuable for gaining a realistic understanding of the project's appearance upon completion and for detecting conflicts at an early stage. Moreover, the 3D viewer provides users with the ability to capture high-quality snapshots from either inside or outside the building. These snapshots may be utilized for documentation, presentations, reporting, or any other relevant purpose.

The second benefit of the intelligent dashboard is its capacity for schedule monitoring and cost control. The success of any construction project, especially large-scale ones, lies in the ability to manage these two factors effectively. Failure to do so may result in schedule delays, cost overruns, or both. Fortunately, Power BI dashboards provide a powerful tool for precise tracking of timelines and budgets. The customized intelligent dashboard plays a crucial part in project management by offering a comprehensive comparison between the actual and the planned progress of the project. This functionality is pivotal for achieving the objectives of schedule monitoring and cost control. The dashboard incorporates two interactively designed 3D models: one representing the planned progress and the other depicting the actual progress to date. The planned 3D model is linked to the project timeline and estimated costs, providing a clear visual representation of the project progression at various stages. The activity table presents all project activities along with their planned start and finish dates. In addition to these models and tables, the dashboard incorporates two-gauge visuals directly linked to the planned 3D model and the activity table. One gauge displays the percentage of the project completed, while the other presents the estimated costs. These gauges serve as quick-reference indicators, providing immediate insights into the project's status. The dashboard also includes a 3D model representing the actual performance of the project, similarly, linked to the activity table and performance visuals.

Throughout the progression of the project, the dashboard dynamically updates to reflect real-time details regarding project progress. The actual 3D model is particularly important, as it depicts completed activities as well as those currently under execution, providing project managers with an accurate representation of accomplished tasks and outstanding work. To enhance usability, the activity table has been customized to have a color-coding system. Completed tasks are marked in green, active ones in orange, and uninitiated ones are uncolored. This visual distinction allows project managers to quickly identify the status of various activities, which facilitates efficient project monitoring.

Furthermore, the gauges provide essential statistics. This information is critical for project managers, as it allows them to assess whether the project is ahead of schedule, behind, or on track, as well as whether it is operating within budgetary constraints or exceeding them. Moreover, by comparing actual progress with planned objectives in real time, project managers can identify potential delays or budget overruns at an early stage, enabling the timely implementation of corrective actions to bring the project back on course. The comparison dashboard is illustrated in Figure 13.

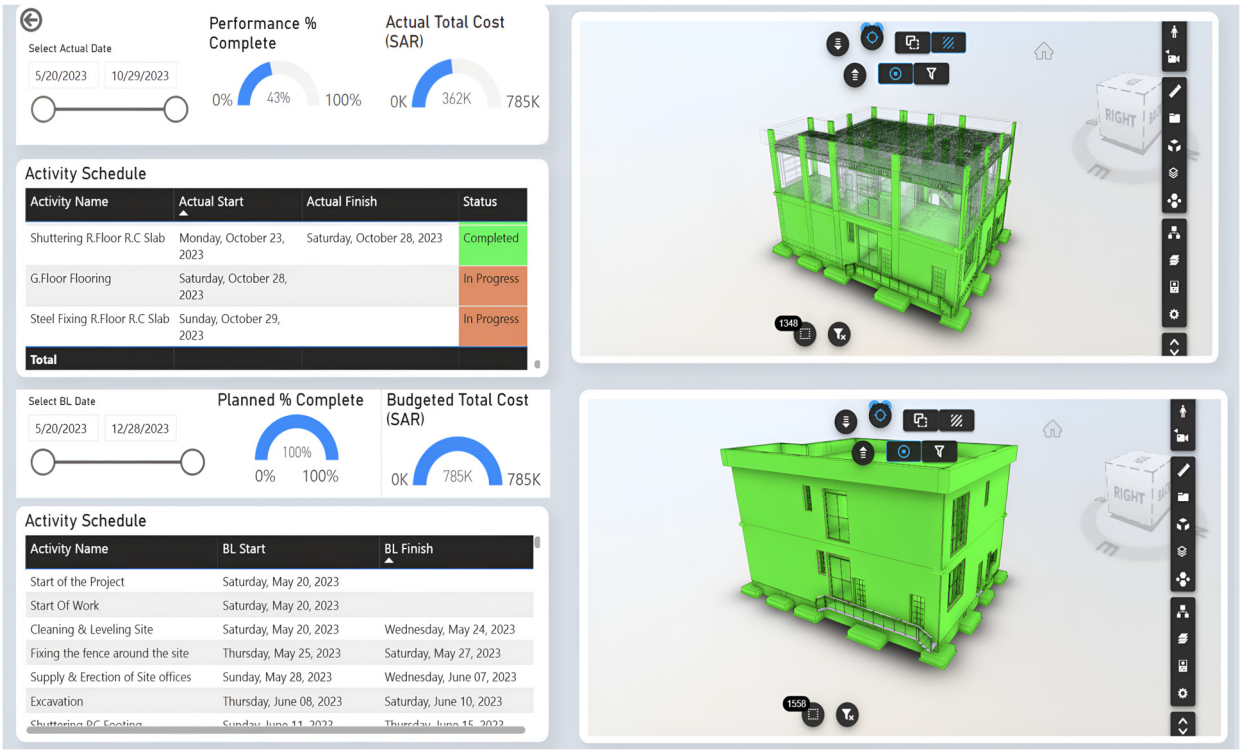


Figure 13. Power BI actual vs. planned progress.

The dashboard further incorporates an interactive Gantt chart linked to the 3D model of the project, as illustrated in Figure 14. This chart serves as a vital tool, providing project managers with immediate visibility into all activities involved within the project. This level of visibility enables stakeholders, especially project managers, to review activity sequences, adjust dates, and analyze relationships between different types of tasks. The clarity of the activity display is further enhanced by organizing activities into groups based on their specific nature, such as earthwork, concrete work, and finishing work. Each group is assigned a distinct color, creating a visually appealing interface that enables stakeholders to readily distinguish between different activity types.

To further enhance the interactivity of the dashboard, a hover feature has been incorporated. When a stakeholder hovers the cursor over any activity, a pop-up appears displaying essential information related to that activity, including the activity duration, start date, end date, percentage complete, name, and other relevant details required for informed decision-making. Additionally, the Gantt chart has been linked to the 3D model of the project. This integration has implemented dynamic interactivity to the dashboard, such as when a stakeholder selects any activity, the 3D model automatically filters to display only the elements associated with the selected activity. This sophisticated integration aids project managers in saving valuable time and accelerates the decision-making process. Furthermore, with both the Gantt chart and the 3D model operating in conjunction, project managers can efficiently assess the impact of changes, identify potential bottlenecks,

and allocate resources more effectively, resulting in a streamlined project workflow and enhanced productivity.

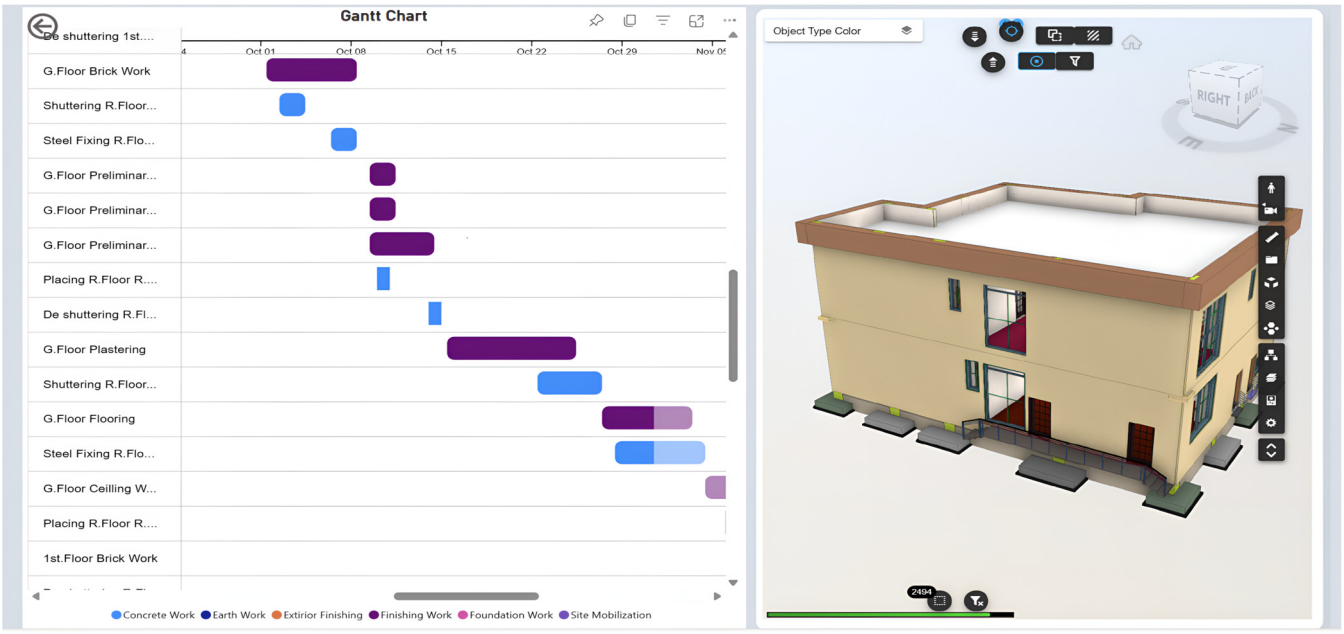


Figure 14. Power BI Gantt dashboard.

The third identified benefit of the intelligent dashboard is its capacity for project performance tracking, which provides a comprehensive view of how various aspects of a project are progressing. Power BI enables stakeholders to establish key performance indicators and monitor them in real time. This feature plays a crucial role in analyzing the efficiency and effectiveness of project execution, while simultaneously fostering a culture of accountability among stakeholders through the provision of a transparent overview of performance metrics. This customized dashboard has contributed to present detailed visualizations pertaining to Earned Value Management (EVM). EVM is a project management technique widely utilized in construction projects to monitor ongoing project performance by measuring the value of work completed against what was originally planned, thereby integrating the dimensions of cost and time into a single, unified performance metric.

EVM measures construction project progress based on three quantitative values: Planned Value (PV), Actual Cost (AC), and Earned Value (EV). These three values play a fundamental role in providing an accurate representation of project status. Planned Value is defined as the authorized budget allocated to the scheduled work up to a specific point in time, serving as a key performance baseline against which actual progress and expenditure can be evaluated. Earned Value refers to the budgeted cost of work completed as of the reporting date. Actual Cost is defined as the actual expenditure incurred for the work performed. Construction managers apply EVM by assigning budgeted costs to each project activity using advanced project management tools such as Primavera P6 or Microsoft Project. These tools facilitate accurate tracking of project activities and expenditures throughout the project lifecycle. Construction teams continuously collect data related to the project progress such as cost through timesheets and invoices. This process ensures that the data used for the calculations of ENM is both recent and accurate. The customized dashboard automates the calculation of PV, EV, and AC by connecting Primavera P6 schedules directly to the dashboard. Consequently, whenever updates are made to the schedule in Primavera, the dashboard immediately recalculates the earned value metrics and displays

them through card and line visuals, providing stakeholders with immediate insights into project performance.

Furthermore, the EVM dashboard utilizes current performance data to generate predictions regarding the final project cost through the calculation of the estimate at completion (EAC). This predictive capability plays an essential role in navigating the complexities of project management by providing a clearer picture of the project's standing relative to its objectives. For instance, if the EAC indicates that the project is likely to exceed its predefined budget or timeline, the project manager can implement corrective actions such as realigning resources, adjusting schedules, or renegotiating the project scope. An additional substantial benefit of the EVM dashboard is its capacity to minimize reliance on subjectivity in performance reporting. Conventional methods of project evaluation tend to incorporate a degree of subjectivity, which can lead to discrepancies and miscommunications. In contrast, EVM provides quantitative indicators that offer a reliable basis for developing reports and evaluating project status. The EVM dashboard serves as an assistant management tool for stakeholders, especially project managers, as it presents an interactive line chart that captures the most important quantitative metrics relevant to project performance throughout the project lifecycle.

The interactive line chart is not only visually appealing, but it also updates in real time, ensuring project managers have access to the most recent data available. With a brief examination of this chart, project managers can rapidly determine whether the project is falling behind schedule, on track, or ahead of the planned schedule. Furthermore, they can readily identify whether the project is exceeding, operating below, or within the allocated budget. To enhance clarity and accessibility for stakeholders who may not possess extensive knowledge of construction management principles, four supplementary metrics have been calculated and are presented on card visuals. These metrics are Schedule Variance (SV), Cost Variance (CV), Schedule Performance Index (SPI), and Cost Performance Index (CPI). A positive SV indicates that the project is progressing ahead of schedule, a negative SV reflects a delay relative to the planned timeline, and a zero SV indicates that the project is precisely on schedule.

Cost Variance follows a similar principle, whereby a positive value indicates that spending is below the budgeted amount, a negative value indicates that the project is over budget, and a zero value indicates that expenditures are aligned with the budget. The efficiency of this dashboard lies in its ability to automate these calculations, sparing stakeholders from the tedious task of manual computation. Furthermore, the dashboard displays the percentage of Cost Variance and Schedule Variance, alerting project managers to the rate of deviation from the planned schedule and authorized budget. For example, the EVM dashboard indicates that the ongoing status of the project is 2.8% lagging behind schedule and 7% over budget. Additionally, the dashboard incorporates predictive capabilities regarding EAC through three distinct scenarios.

The first scenario evaluates the situation in which no corrective actions have been implemented, maintaining the current Cost Performance Index for the remainder of the project. Secondly, it considers a scenario where corrective actions are taken, and the remaining work is completed at the planned rate. Finally, it assesses a situation where both the Cost Performance Index and the Schedule Performance Index influence the completion of the remaining work. Furthermore, the dashboard automatically calculates the projected profit or loss for each scenario and presents these figures in card visuals, enabling project managers to understand the financial implications of each potential outcome. This dashboard empowers stakeholders, especially project managers with the ability to monitor the critical key performance indicators (KPIs) of the project, eliminating the need for manual

calculations or reliance on traditional reporting methods. The EVM dashboard is illustrated in Figure 15.

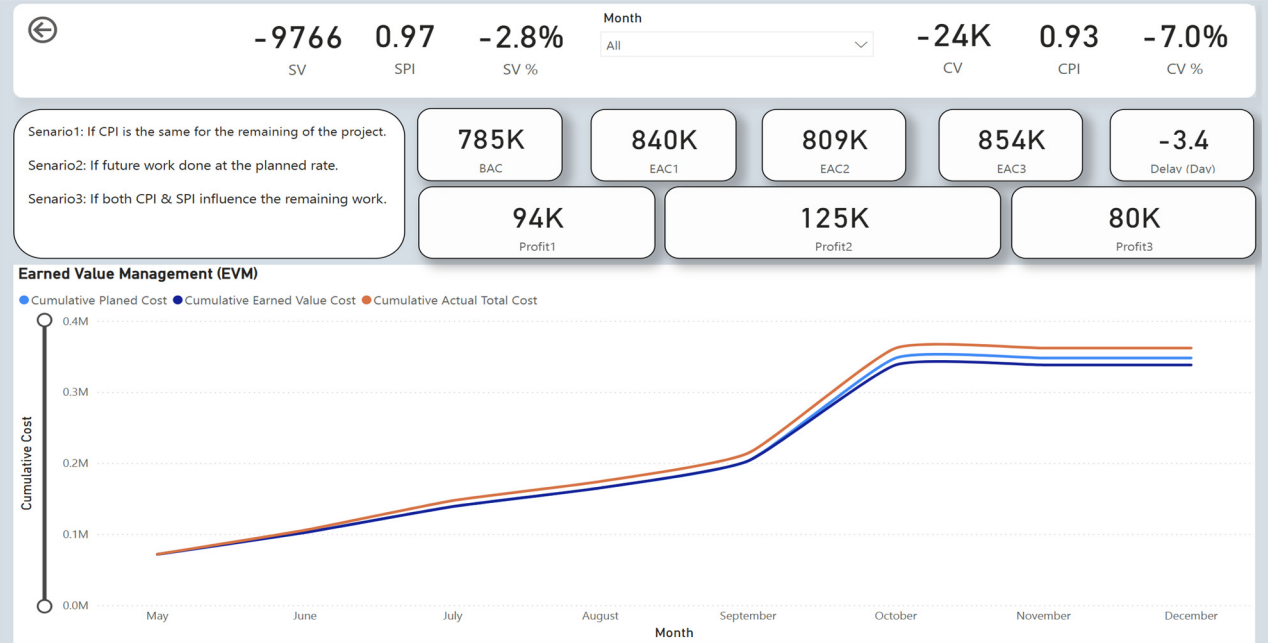


Figure 15. Power BI EVM dashboard.

Enhanced collaboration and communication among stakeholders represent the fourth benefit that customized Power BI dashboards offer. Power BI facilitates the seamless sharing of insights and reports between team members, engineers, clients, project managers, contractors, and other stakeholders, transforming traditional methods of data sharing into a more dynamic and interactive process. This feature is extremely important in today's dynamic business environment, where the availability of accurate and timely information can make all the difference in decision-making processes. The sharing of reports and insights can be readily accomplished through the Power BI workspace. This workspace is not only user-friendly but also easily accessible, enabling users to access it via laptops, web browsers, or smartphones. This level of accessibility eliminates the need for the installation of complex software or extensive technical expertise. Furthermore, the data shared within this platform is protected by a high level of encryption, underpinned by Microsoft's advanced security mechanisms. whereas data is secured during transmission through Transport Layer Security (TLS) and during storage through Azure Storage encryption. These security mechanisms offer significant assurance that sensitive information remains confidential, giving stakeholders confidence when collaborating and sharing information. In addition, Power BI supports the assignment of defined roles to each stakeholder within the workspace. These roles are crucial, as they delineate the extent of access and the range of activities each stakeholder may undertake within the workspace, thereby establishing a structured and organized environment.

5. Conclusions and Recommendations

5.1. Conclusions

Despite the widespread adoption of Business Intelligence tools across various sectors, their application within the construction management sector remains limited. This study presented the development of a customized Power BI dashboard that utilizes Building Information Modeling to enhance real-time monitoring, decision-making, and control of construction projects. This approach has been illustrated throughout a detailed case study

illustrating its practical application. By integrating BIM data with the analytical capabilities of BI, the customized dashboard addresses the existing gaps in traditional project management practices by providing dynamic and interactive visualizations of key performance indicators of the project along with a convenient platform that promotes stakeholders' communication and collaboration. This dashboard can automate the extraction of data from multiple sources such as geometry and properties from Autodesk service platform and schedule and cost from Primavera. A connection between the dashboard and the data sources is established once, after which the dashboard automatically refreshes and updates each time the data sources are modified. This automatic update will provide stakeholders with the most updated information regarding project progress. Furthermore, the dashboard comprises multiple pages, each customized to serve a specific purpose. For example, the planned versus actual dashboard has been customized to facilitate schedule monitoring and cost control. In conclusion, this study has demonstrated that the integration of Building Information Modeling with Business Intelligence extends beyond a theoretical concept, representing a practical and transformative tool for the management of construction projects. As construction projects continue to grow in complexity, adopting Business Intelligence tools such as Power BI dashboards will be critical to address these challenges and stay competitive in the market.

5.2. Recommendations for Future Work

Future work could explore the integration of Power BI dashboards with advanced technologies such as mixed reality. This integration would allow project stakeholders to visualize and overlay 3D rich data models directly onto construction sites. This capability would enable users to compare actual progress against planned, detect deviations in real-time, and interact with virtual components of the model, while simultaneously monitoring key performance indicators of the project. Such integration has the potential to transform decision-making processes in the construction sector. Future work may also consider extending the Power BI framework to incorporate risk assessment and management tools. Construction projects naturally involve unpredictability, where they are often influenced by numerous risk factors such as cost fluctuations, schedule changes, safety hazards, environmental conditions, and resource availability. Future dashboards could incorporate structured risk assessment frameworks that classify, quantify, and track these risk factors throughout the project lifecycle. Additionally, integrating Monte Carlo simulation outputs with Power BI dashboards could display probability distributions, risk curves, sensitivity analyses, and confidence intervals for cost and schedule outcomes. This would enable project managers to understand the range of possible scenarios and make well-informed contingency planning decisions.

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References

1. Hossain, M.A.; Nadeem, A. Towards digitizing the construction industry: State of the art of construction 4.0. *Proc. Int. Struct. Eng. Constr.* **2019**, *10*, 1–6. [CrossRef]
2. Nikolić, M.; Cerić, A. Classification of key elements of construction project complexity from the contractor perspective. *Buildings* **2022**, *12*, 696. [CrossRef]
3. Adel, L. The Benefits of Power BI—PMO365: Project Portfolio Management Solution. Pmo365. 2023. Available online: https://pmo365.com/blog/top-9-best-power-bi-capabilities (accessed on 18 March 2026).
4. McKinsey & Company. The Next Normal in Construction: How Disruption is Reshaping the World's Largest Ecosystem. 2020. Available online: https://www.mckinsey.com/capabilities/operations/our-insights/the-next-normal-in-construction-how-disruption-is-reshaping-the-worlds-largest-ecosystem#/ (accessed on 18 March 2026).
5. Yang, Z.W.; Guan, G.; Fang, H.; Xue, X. Average propagation length analysis for the change trend of China's construction industry chain. *J. Asian Archit. Build. Eng.* **2021**, *21*, 1078–1092. [CrossRef]
6. Imafidon, H.; Enwerem, M.; Boye, A. Adapting Green Building Practices and Smart Technology in Developing Countries. *Afr. J. Environ. Sci. Renew. Energy* **2024**, *16*, 183–202. [CrossRef]
7. Malik, A.N.; Mbewe, P.B.K.; Kavishe, N.; Mkandawire, T.; Adetoro, P.E. Sustainable Construction Practices in Building Infrastructure Projects: The Extent of Implementation and Drivers in Malawi. *Sustainability* **2024**, *16*, 10825. [CrossRef]
8. Al-Raqeb, H.; Ghaffar, S.H. The Role of BIM 6D and 7D in Enhancing Sustainable Construction Practices: A Qualitative Study. *Technologies* **2025**, *13*, 65. [CrossRef]
9. Cristóbal, J.R.S.; Carral, L.; Diaz, E.; Fraguela, J.A.; Iglesias, G. Complexity and Project Management: A General Overview, Complexity. 2018. Available online: https://www.hindawi.com/journals/complexity/2018/4891286/ (accessed on 18 March 2026).
10. Alsharef, A.; Ovid, A.; Uddin, S.M.J.; Albert, A. Biggest Challenges Facing the Construction Industry. In Proceedings of the Construction Research Congress 2024, Des Moines, IA, USA, 20–23 March 2024; pp. 652–660. [CrossRef]
11. Asif, M. Growth and sustainability trends in the buildings sector in the GCC region with particular reference to the KSA and UAE. *Renew. Sustain. Energy Rev.* **2016**, *55*, 1267–1273. [CrossRef]
12. Asif, M.; Dehwah, A.H.; Ashraf, F.; Khan, H.S.; Shaukat, M.M.; Hassan, M. Life cycle assessment of a three-bedroom house in Saudi Arabia. *Environments* **2017**, *4*, 52. [CrossRef]
13. Karel. Vision 2030: Expansion of Construction Sector in KSA. GOGlobe. 2024. Available online: https://www.go-globe.com/vision-2030-and-expansion-of-construction-sector/ (accessed on 18 March 2026).
14. Alshammari, A.; Ghazali, F.E.M. A Comprehensive Review of the Factors and Strategies to Mitigate Construction Projects Delays in Saudi Arabia. *Open Constr. Build. Technol. J.* **2024**, *18*. [CrossRef]
15. Ahmad, S.; Aftab, F.; Eltayeb, T.K.; Siddiqui, K.A. Identifying critical success factors for construction projects in Saudi Arabia. *E3S Web Conf.* **2023**, *371*, 02047. [CrossRef]
16. Alhassan, S.A.; Almanei, M.; Salonitis, K. Challenges in Implementing Lean and Green in Saudi Construction Sector. In *Advances in Manufacturing Technology XXXVI*; IOS Press: Amsterdam, The Netherlands, 2023. [CrossRef]
17. Abougamil, R.A.; Thorpe, D.; Heravi, A. A BIM Package with NEC4 Contract Option to Mitigate Construction Disputes in the Kingdom of Saudi Arabia. *Buildings* **2024**, *14*, 2009. [CrossRef]
18. Alhamoudi, A.M. Enabling Industry 4.0 Technologies to Drive Sustainable Housing Delivery across the Housing Supply Value Chain in Saudi Arabia: Challenges and Prospects. *Sustainability* **2024**, *16*, 5413. [CrossRef]
19. Adhikari, S.; Roca, A. Impact of BIM technology on the construction project management process. *Epic. Ser. Built Environ.* **2024**, *5*, 948–956. [CrossRef]
20. Cholakis, P. Definition of BIM—Building Information Modeling—Defintions from Various Sources. LEAN Construction Project Delivery Methods—Job Order Contracting. 2012. Available online: https://buildinginformationmanagement.wordpress.com (accessed on 18 March 2026).
21. Mi, Z.; Li, J. Maximizing project efficiency and collaboration in construction management through building information modeling (BIM). *Appl. Comput. Eng.* **2024**, *72*, 24–29. [CrossRef]
22. Igba, E.; Danquah, E.O.; Ukpoju, E.A.; Obasa, J.; Olola, T.M.; Enyejo, J.O. Use of Building Information Modeling (BIM) to improve construction management in the USA. *World J. Adv. Res. Rev.* **2024**, *23*, 1799–1813. [CrossRef]
23. Hnin, T. Understanding BIM (Building Information Modeling): The Complete Guide. Novatr.com. 2022. Available online: https://www.novatr.com/blog/detailed-bim-guide (accessed on 18 March 2026).
24. Frankenfield, J. What is Business Intelligence (BI)? Types, Benefits, and Examples. Investopedia. 2024. Available online: https://www.investopedia.com/terms/b/business-intelligence-bi.asp (accessed on 18 March 2026).
25. Biswal, A. What is Power BI?: Architecture, and Features Explained. Simplilearn.com. 2025. Available online: https://shorturl.at/S6TsM (accessed on 18 March 2026).
26. Iseminger, D. Data sources in Power BI Desktop—Power BI. Microsoft Learn. 2024. Available online: https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-data-sources (accessed on 18 March 2026).

27. Rodrigues, F.; Alves, A.D.; Matos, R. Construction Management supported by BIM and a business intelligence tool. *Energies* **2022**, *15*, 3412. [CrossRef]
28. Lopes, A.; Boscarioli, C. Business intelligence and analytics to support management in construction: A systematic literature review. *Rev. Bras. Comput. Apl.* **2020**, *13*, 27–41. [CrossRef]

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